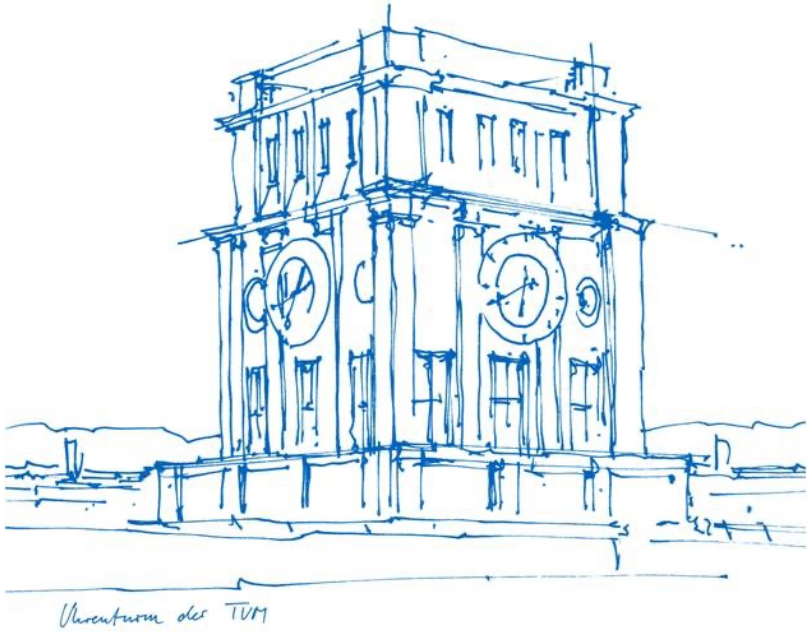
Energy Informatics Lab

Assignment 4: Optimal Power Flow

Technical University of Munich

TUM School of Engineering and Design

Professorship of Energy Management Technologies



Today‘s Agenda

- Background on optimal power flow
- Introduction of assignment

Today‘s Agenda

- **Background on optimal power flow**
- Introduction of assignment

Background: Economic Dispatch

- Objective: minimize the cost of generation
- Constraints
  - Equality Constraint: load generation balance
  - Inequality constraints: upper and lower limits on for generation
- Example:

$$min \sum_{i \in (G)} C_i\left(P_i^G\right)$$

s.t.

$$\sum P_i^G = \sum P_i^L$$

$$P_i^{G,min} \leq P_i^G \leq P_i^{G,max} \quad \forall i \in G$$

Economic Dispatch Limitations

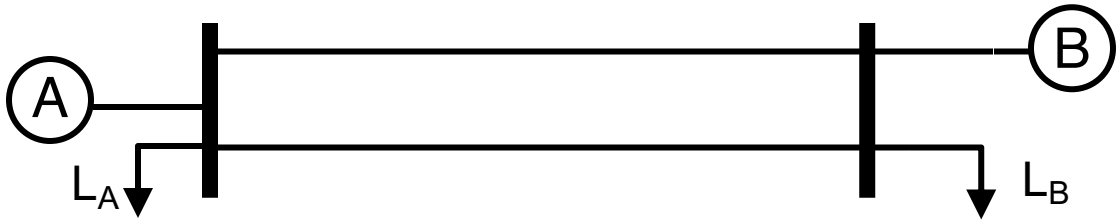
- Generating units and loads are not all connected to the same bus
- May result in unacceptable flows or voltages in the network

Example

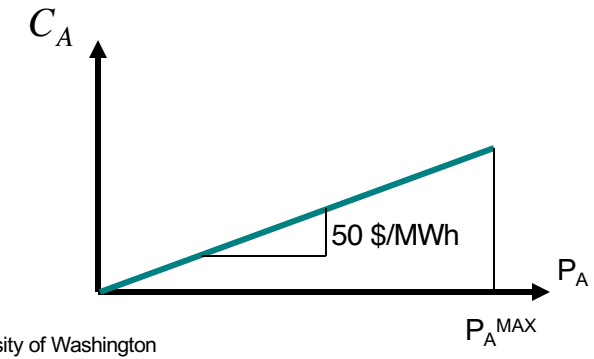
- The following slides are taking from an Optimal Power Flow lecture from Daniel Kirschen of the University of Washington

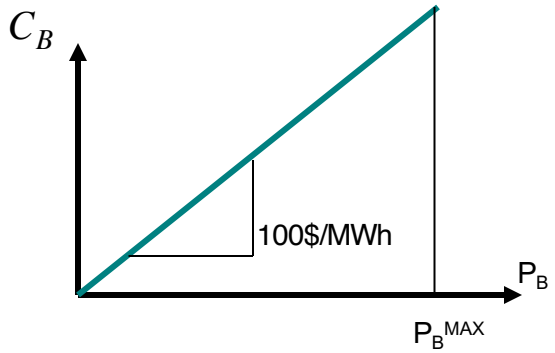
- https://www.youtube.com/watch?v=miNer-pRIe8

Example of network limitation



Maximum flow on each line: 100MW

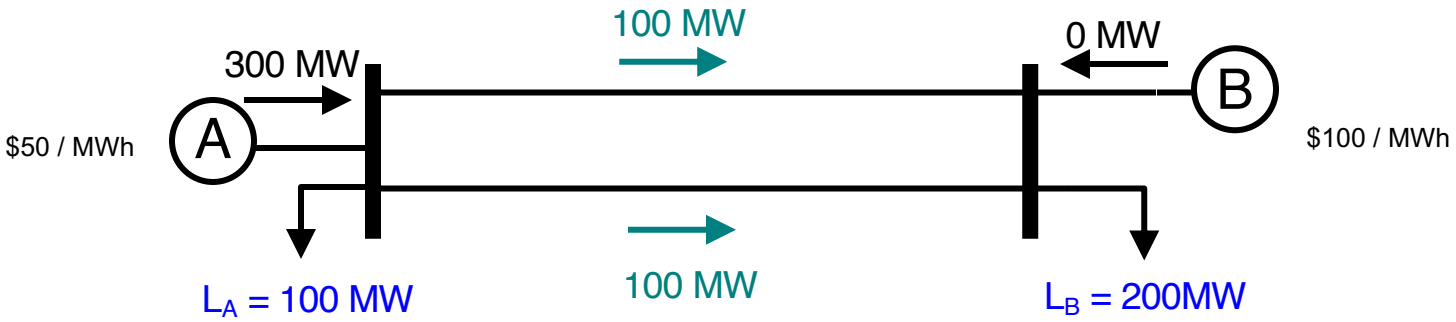




© 2011 D. Kirschen and the University of Washington

Acceptable ED solution

Maximum flow on each line: 100MW



The solution of this (trivial) economic dispatch is:

$$P_A = 300 \text{ MW}$$

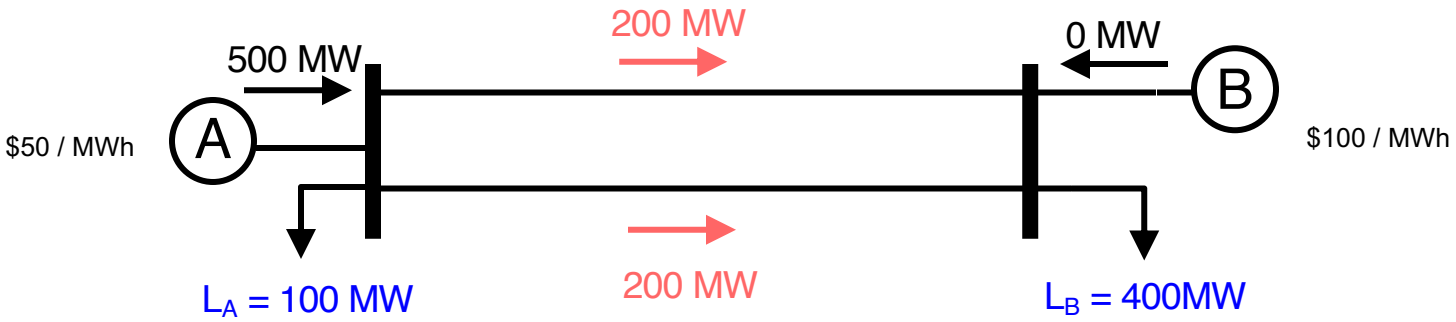
$$P_B = 0 \text{ MW}$$

The flows on the lines are below the limit
The economic dispatch solution is acceptable

© 2011 D. Kirschen and the University of Washington

Unacceptable ED solution

Maximum flow on each line: 100MW



The solution of the economic dispatch is:

$$P_A = 500 \text{ MW}$$

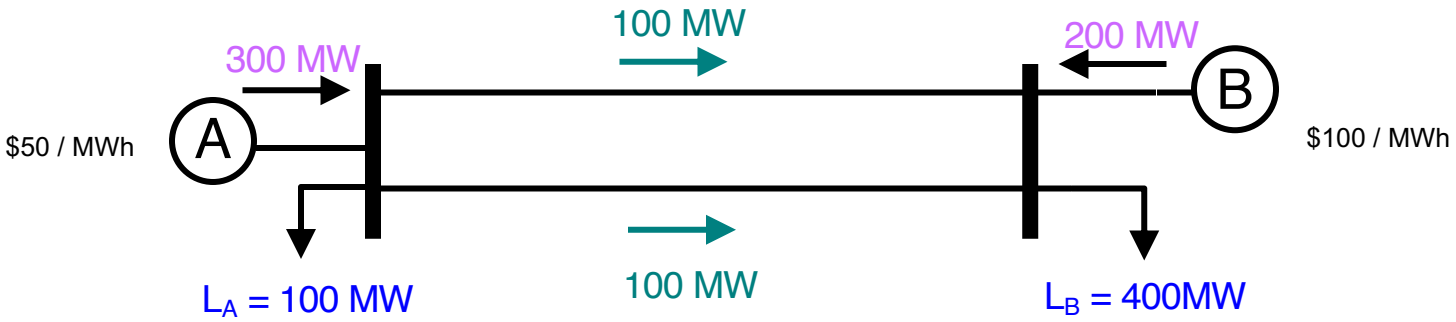
$$P_B = 0 \text{ MW}$$

The resulting flows exceed their limit
The economic dispatch solution is not acceptable

© 2011 D. Kirschen and the University of Washington

Modified ED solution

Maximum flow on each line: 100MW



In this simple case, the solution of the economic dispatch can be modified easily to produce acceptable flows.

This could be done mathematically by adding the following inequality constraint: $P_A - L_A \leq 200\text{MW}$

However, adding inequality constraints for each problem is not practical in more complex situations
We need a more general approach

© 2011 D. Kirschen and the University of Washington

Optimal Power Flow

- Power optimization problem that takes the systems physical characteristics into account

- This includes:
  - Power balances at buses
  - Network operating limits (line flows and voltages)
  - Limits on control variables

- Complex optimization that needs to be solved very frequently and in real time

Optimal Power Flow

- Objective Function:
  - Minimize generating cost

$$\min_{u} \sum_{i=1}^{N_G} C_i(P_i)$$

- Equality constraints:
  - Active and reactive power balance

$$P_k^G - P_k^L = \sum_{i=1}^{N} V_k V_i [G_{ki} \cos(\theta_k - \theta_i) + B_{ki} \sin(\theta_k - \theta_i)]$$

$$Q_k^G - Q_k^L = \sum_{i=1}^{N} V_k V_i [G_{ki} \sin(\theta_k - \theta_i) - B_{ki} \cos(\theta_k - \theta_i)]$$

$$k = 1, \ldots N$$

- Inequality Constraints:
  - Control variables
  - Limits on flows
  - Limits on voltages

$$\underline{u} \leq u \leq \overline{u}$$

$$\left| F_{ij} \right| \leq \overline{F_{ij}}$$

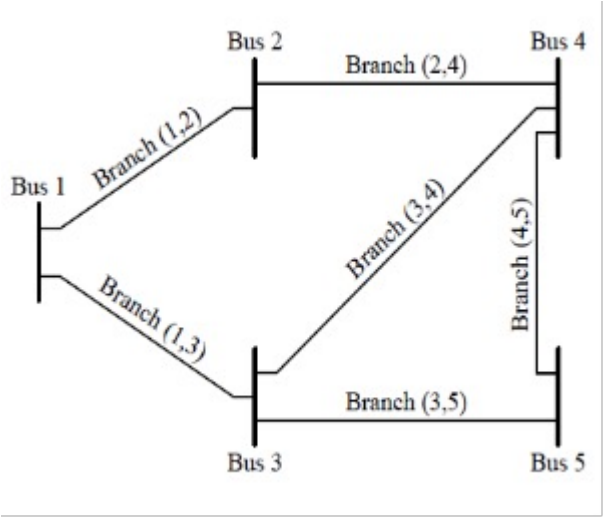
$$\underline{V_j} \leq V_j \leq \overline{V_j}$$

Today‘s Agenda

- Background on optimal power flow
- **Introduction of assignment**

Optimal Power Flow

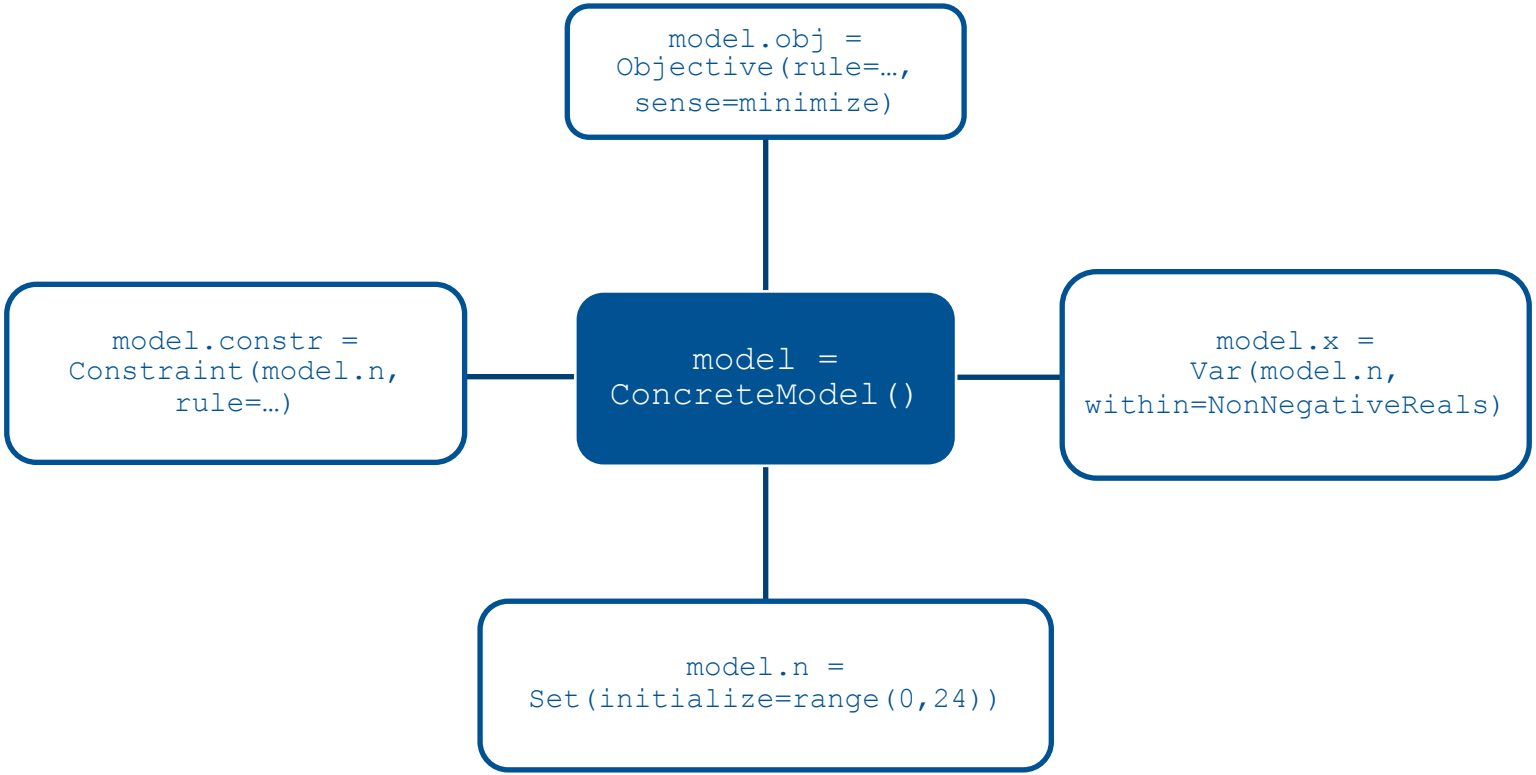
- Implement the AC and DC power flow optimization problems for an example network



Optimal Power Flow

- Using pyomo and the NEOS server
- Starter code in opf_starter.ipynb
- Utility functions in opf_utils.py
- Network / system data in OPFData.xlsx

Optimization in Pyomo



https://pyomo.readthedocs.io/en/stable/index.html

Schedule

# Meeting	Location	Content
1st Session	N3815	Introduction of assignment; work on tasks
Until 1st Session of next assignment	N3815	Work on task; presentation
Until 2nd Session of next assignment	Home	Work on report